



Advanced Linear Algebra: Foundations to Frontiers

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1.2.2 What is a vector norm? 📌

fit width

A vector norm extends the notion of an absolute value to vectors. It allows us to measure the magnitude (or length) of a vector. In different situations, a different measure may be more appropriate.

Definition 1.2.2.1. Vector norm. Let $\nu : \mathbb{C}^m \rightarrow \mathbb{R}$. Then ν is a (vector) norm if for all $x, y \in \mathbb{C}^m$ and all $\alpha \in \mathbb{C}$

- $x \neq 0 \Rightarrow \nu(x) > 0$ (ν is positive definite),
- $\nu(\alpha x) = |\alpha| \nu(x)$ (ν is homogeneous), and
- $\nu(x + y) \leq \nu(x) + \nu(y)$ (ν obeys the triangle inequality).

Homework 1.2.2.1. TRUE/FALSE: If $\nu : \mathbb{C}^m \rightarrow \mathbb{R}$ is a norm, then $\nu(0) = 0$.

► [Hint](#) ► [Answer](#) ▼ [Solution](#)

Let $x \in \mathbb{C}^m$ and, just for clarity this first time, $\vec{0}$ be the zero vector of size m so that 0 is the scalar zero. Then

$$\begin{aligned}
& \nu(\vec{0}) \\
&= \langle 0 \cdot x = \vec{0} \rangle \\
& \nu(0 \cdot x) \\
&= \langle \nu(\dots) \text{ is homogeneous} \rangle \\
& 0\nu(x) \\
&= \langle \text{algebra} \rangle \\
& 0
\end{aligned}$$

Remark 1.2.2.2. We typically use $\|\cdot\|$ instead of $\nu(\cdot)$ for a function that is a norm.